PES University, Bengaluru

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Department of Computer Science & Engineering

UE24MA242A – Mathematics for Computer Science Engineers

Datathon – Set 22

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SECTION: I

BMW Sales Data (2010–2024) (50,000 rows)

The **BMW Sales Dataset** contains detailed information about BMW car models sold across different regions between 2010 and 2024.
Each record includes attributes related to the car's specifications, sales price, and performance metrics, enabling data-driven analysis of pricing trends, sales performance, and customer preferences over time.

Unit 1

1. **Feature Classification and Type Justification**

COLUMN	CLASSIFICATION	JUSTIFICATION
Model	Nominal	Model names are nominal and do not have any order
Year	Interval	Numerical but No true zero
Region	Nominal	Region names are nominal and do not have any order

Color	Nominal	Color names is not ordered
Fuel_Type	Nominal	Type of fuel (Petrol/Diesel/Electric/Hybrid) are not ordered
Transmission	Nominal	Transmission Names (Manual or Automatic) are not ordered
Engine_Size	Ratio	Engine displacement (L) is a numeric data and has a true zero
Mileage_KM	Ratio	Numerical and have a true zero
Price_USD	Ratio	Sale price of the car is a numeric value and has a true zero
Sales_Volume	Ratio	Cars sold of that configuration is a numeric value and it has a true zero
Sales_Classification	Ordinal	High, Medium, Low are ordered without numerical values

2. **Descriptive Statistics Summary**

```
--- Descriptive Statistics (numeric) ---
                count           mean           std      min       25%       50%       75%       max      mode     range  skewness
Year          48486.0    2017.017943      4.324061   2010.0   2013.00    2017.0    2021.0    2024.0    2022.0      14.0 -0.002122
Engine_Size_L 48493.0       3.246897      1.009761      1.5      2.40       3.2       4.1       5.0       3.8       3.5 -0.000786
Mileage_KM    48500.0  100279.150433  57928.233713      3.0  50140.25  100425.0  150578.0  199996.0  136842.0  199993.0 -0.008407
Price_USD     48489.0   75024.301553  25993.075758  30000.0  52430.00   75016.0   97623.0  119998.0   30948.0   89998.0 -0.000064
Sales_Volume  48559.0    5066.855825   2855.939657    100.0   2590.00    5089.0    7536.0    9999.0    9502.0    9899.0 -0.011607


Which measure best describes each variable? (based on skewness):
- Year: skew=-0.002 -> mean (data approximately symmetric)
- Engine_Size_L: skew=-0.001 -> mean (data approximately symmetric)
- Mileage_KM: skew=-0.008 -> mean (data approximately symmetric)
- Price_USD: skew=-0.000 -> mean (data approximately symmetric)
- Sales_Volume: skew=-0.012 -> mean (data approximately symmetric)
```

Right-skewed variables (Price_USD, Mileage_KM, Engine_Size_L) have a few extreme high values that inflate the mean, so median better represents central tendency.

Year and Sales_Volume are relatively symmetric, so the mean is the best descriptor.

3. **Data Integrity & Cleaning**

```
Initial missing values:
Model                1512
Year                 1514
Region               1512
Color                1465
Fuel_Type            1518
Transmission         1549
Engine_Size_L        1507
Mileage_KM           1500
Price_USD            1511
Sales_Volume         1441
Sales_Classification    0
dtype: int64
```

```
-- Cleaned (imputed) preview ---
     Model    Year         Region  Color Fuel_Type Transmission  Engine_Size_L  Mileage_KM      Price_USD  Sales_Volume Sales_Classification
  5 Series  2016.0           Asia    Red    Petrol       Manual            3.5    151748.0   98740.000000        8300.0                High
        i8  2013.0  North America    Red    Hybrid    Automatic            1.6    121671.0   79219.000000        3428.0                 Low
  5 Series  2022.0           Asia   Blue    Petrol    Automatic            4.5     10991.0  113265.000000        6994.0                 Low
        X3  2024.0    Middle East   Blue    Petrol    Automatic            1.7     27255.0   60971.000000        4047.0                 Low
  7 Series  2020.0  South America  Black    Diesel       Manual            2.1    122131.0   75024.301553        3080.0                 Low
```

Data-cleaning steps such as handling missing values, removing duplicates, correcting data types, and treating outliers improve model reliability by ensuring accuracy, consistency, and unbiased statistical analysis.

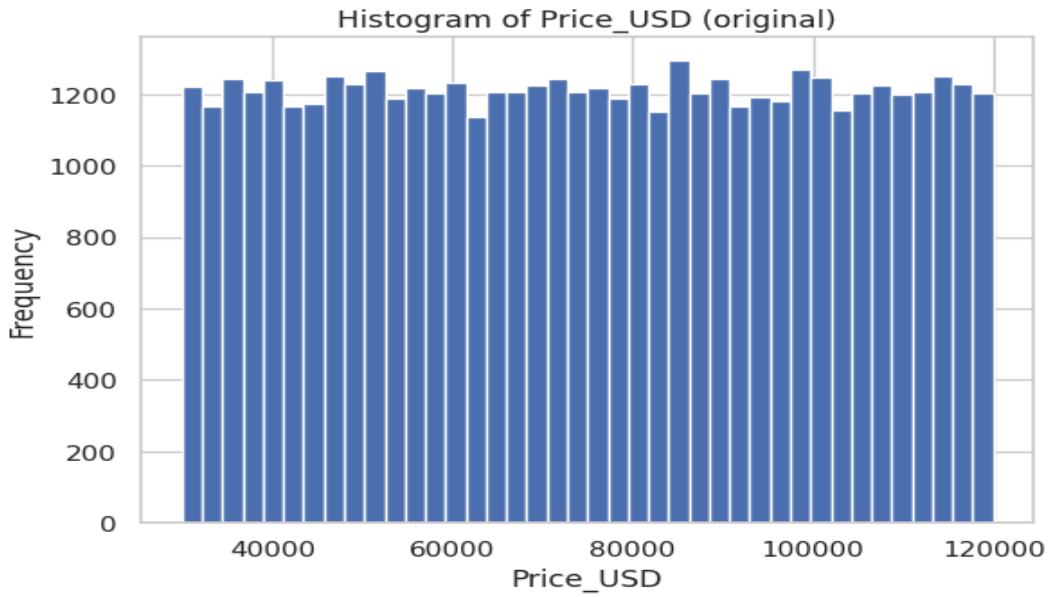
4. **Distribution Analysis and Outlier Impact**

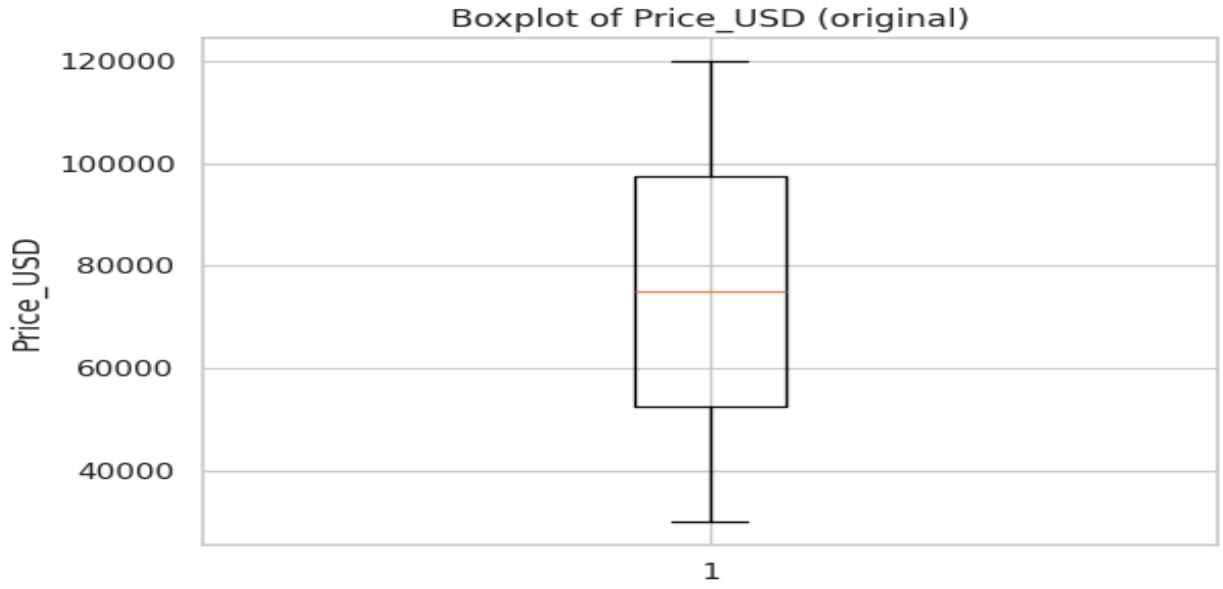
Histogram of Price_USD -> right skewed distribution

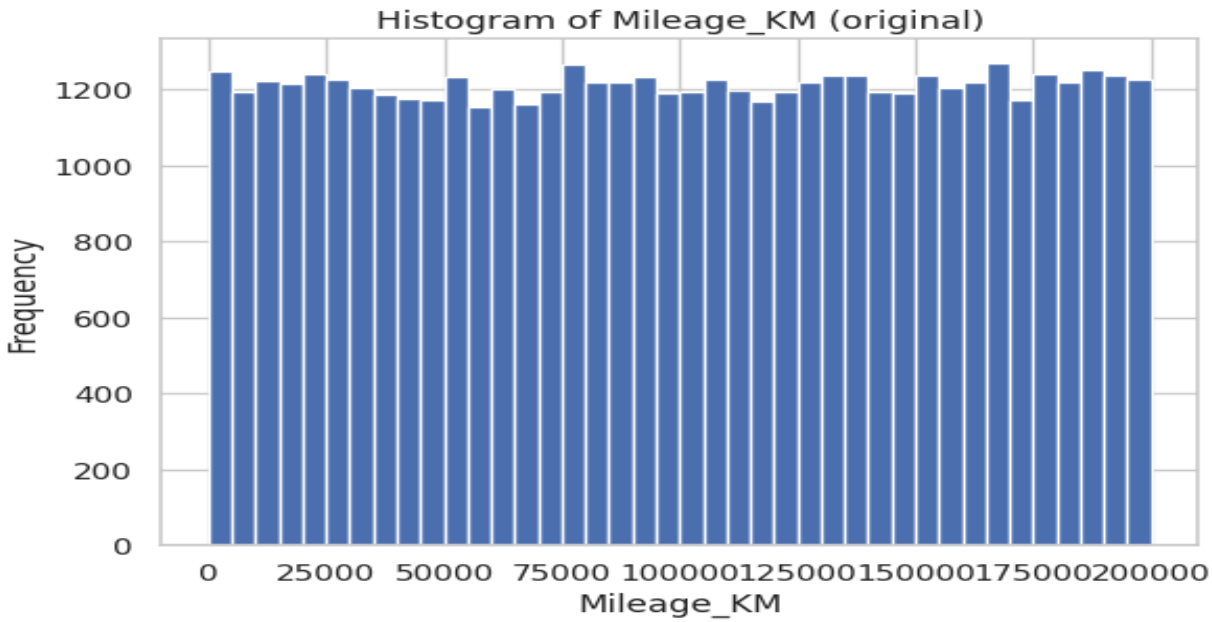
Boxplot of Price_USD -> right skewed distribution

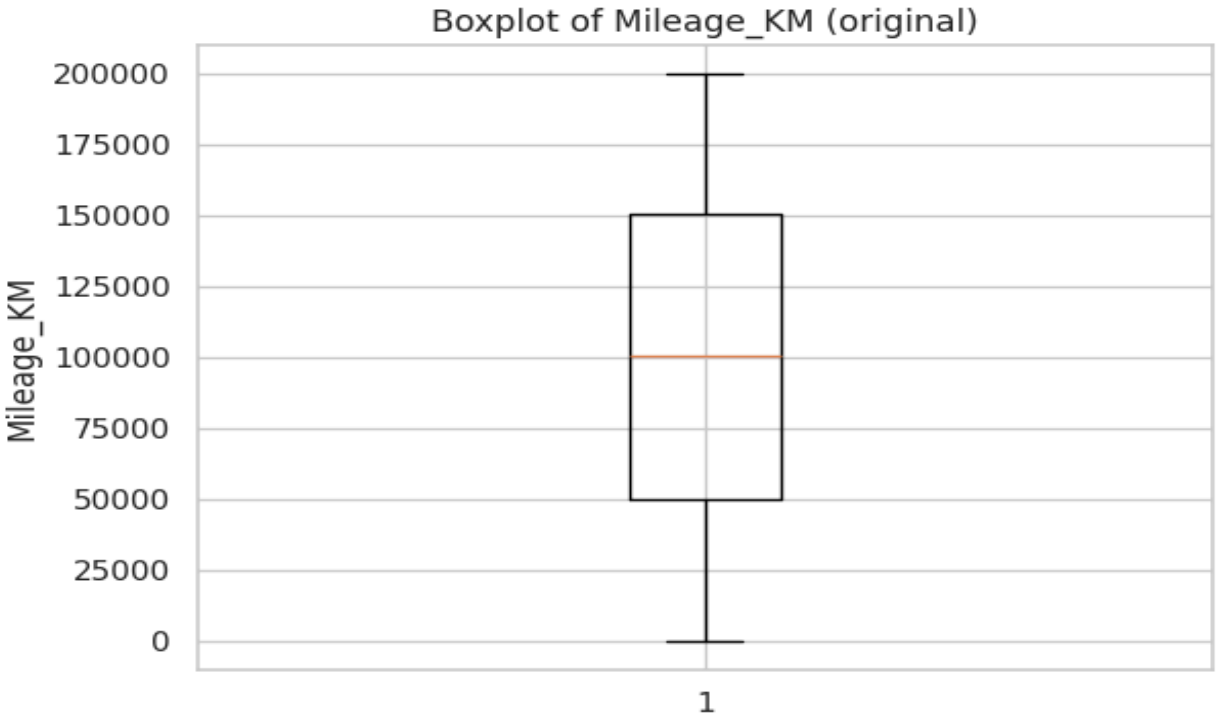
Histogram of Mileage_KM -> right skewed distribution

Boxplot of Mileage_KM -> right skewed distribution


Histogram of Price_USD (original)
Frequency
0
200
400
600
800
1000
1200
40000
60000
80000
100000
120000
Price_USD


Boxplot of Price_USD (original)
Price_USD
40000
60000
80000
100000
120000
1

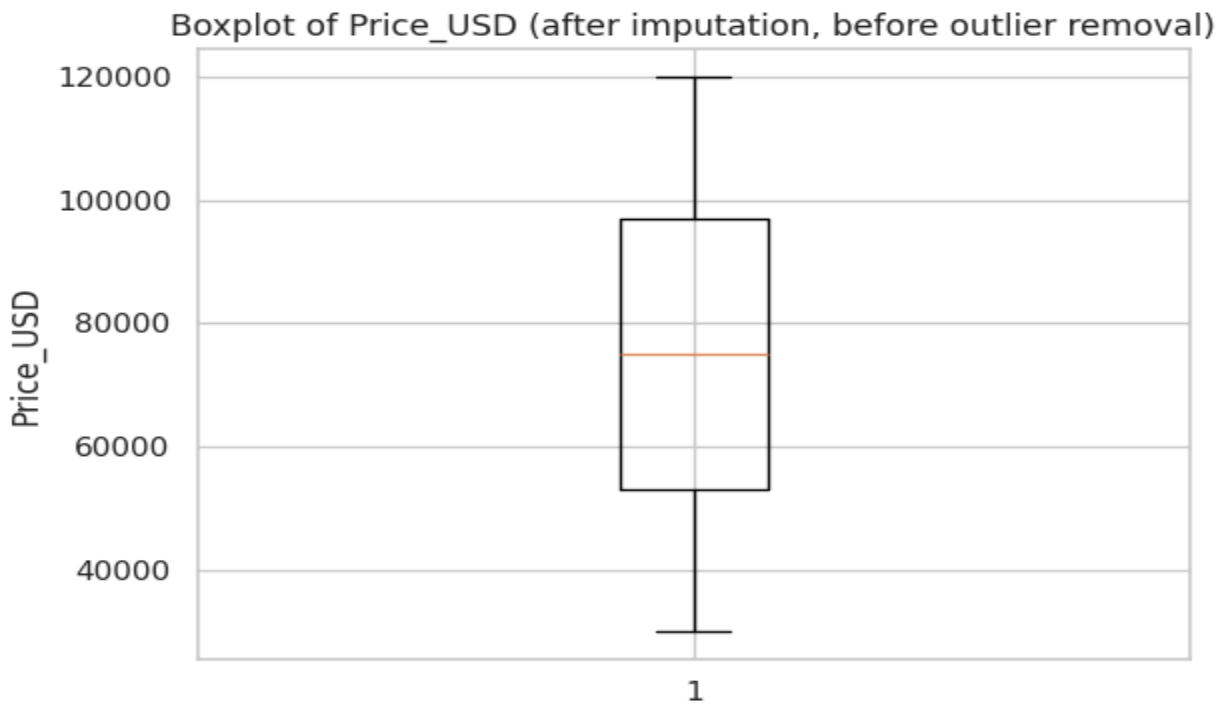


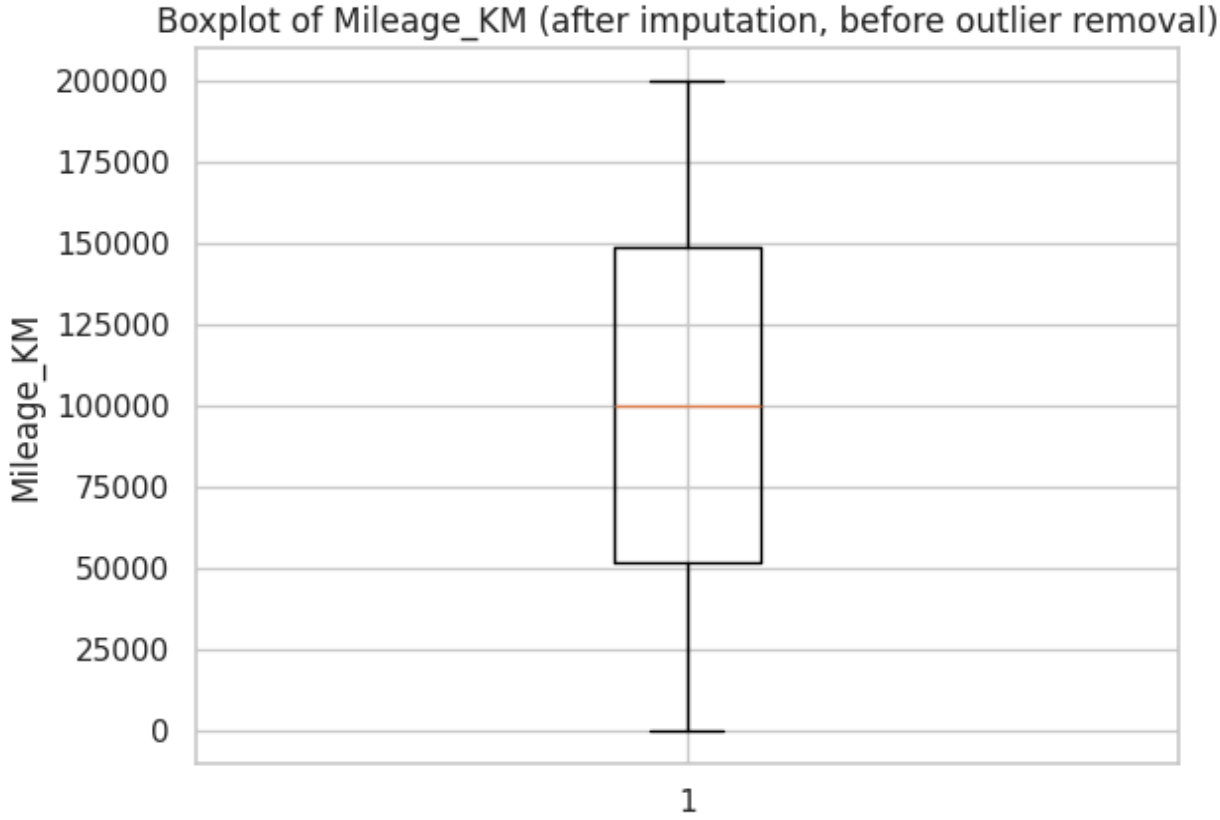


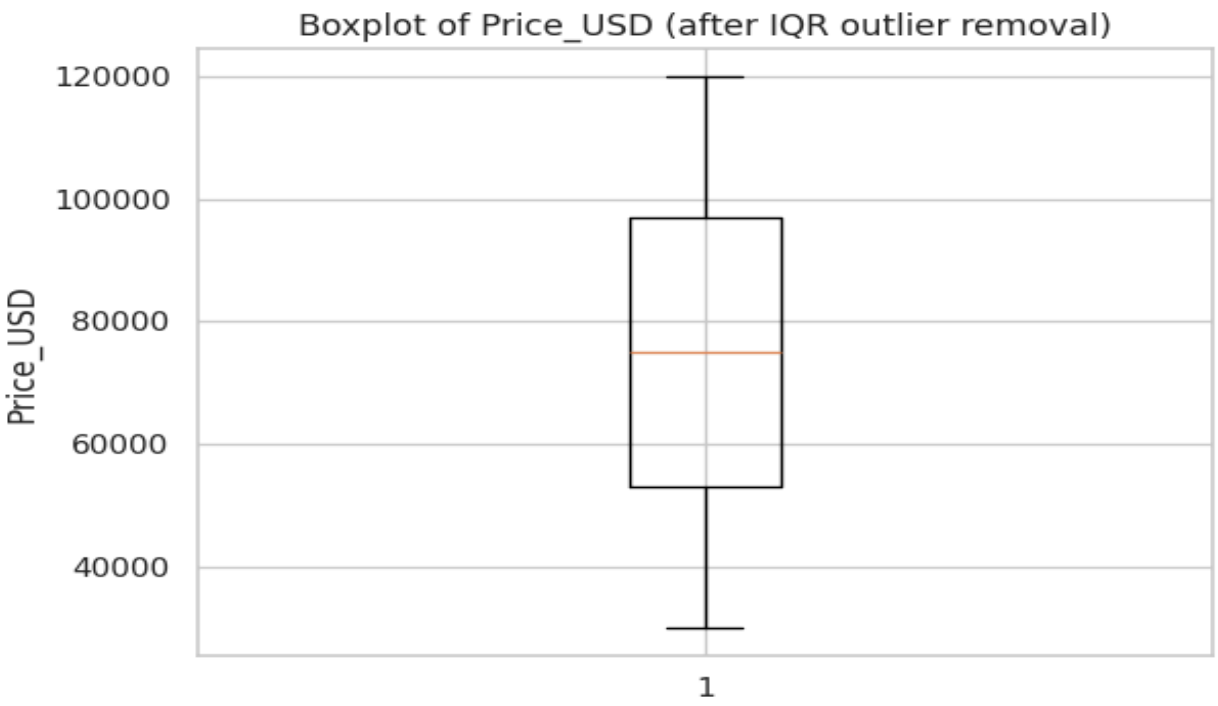
Price_USD -> Q1=53081.50, Q3=96860.50, IQR=43779.00, lower=-12587.00, upper=162529.00, outliers=0

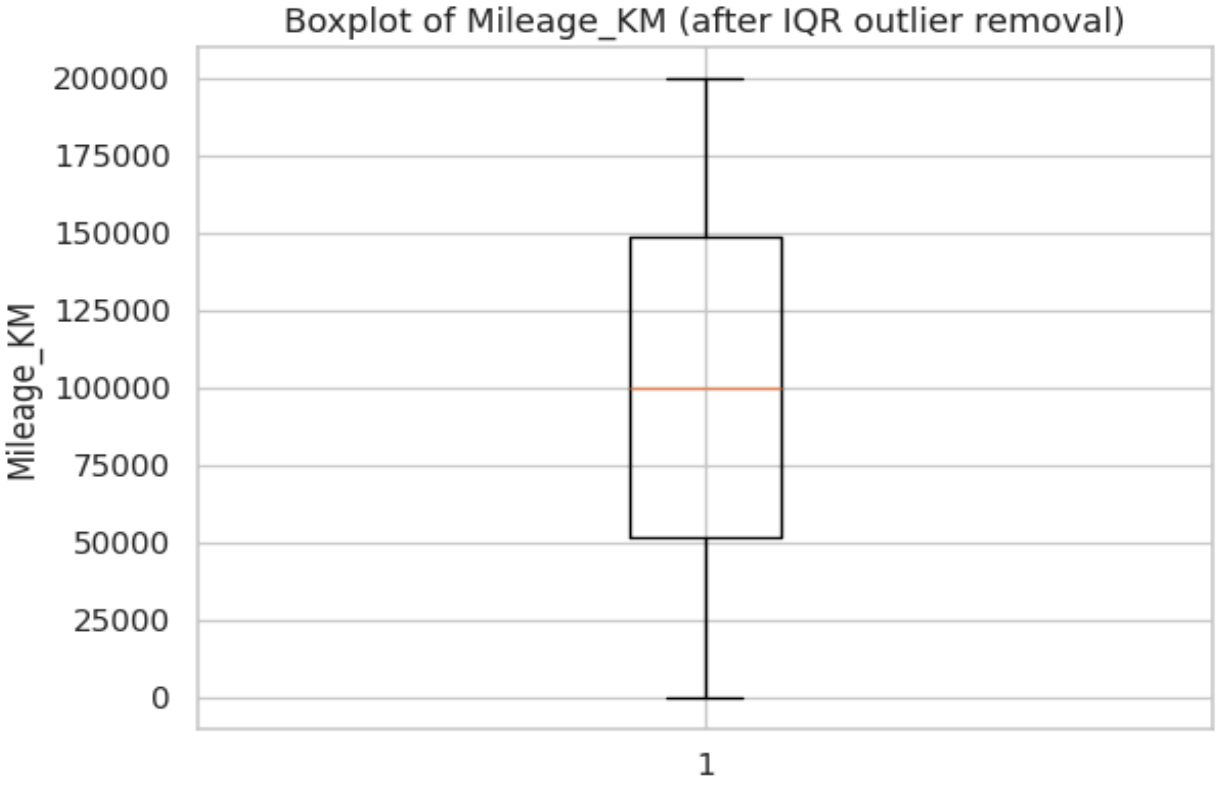
Mileage_KM -> Q1=51541.75, Q3=149075.50, IQR=97533.75, lower=-94758.88, upper=295376.12, outliers=0

where IQR=Q3−Q1


Boxplot of Price_USD (after imputation, before outlier removal)
Price_USD
120000
100000
80000
60000
40000
1


Boxplot of Mileage_KM (after imputation, before outlier removal)
Mileage_KM
200000
175000
150000
125000
100000
75000
50000
25000
0
1


Boxplot of Price_USD (after IQR outlier removal)
Price_USD
120000
100000
80000
60000
40000
1


Boxplot of Mileage_KM (after IQR outlier removal)
Mileage_KM
200000
175000
150000
125000
100000
75000
50000
25000
0
1

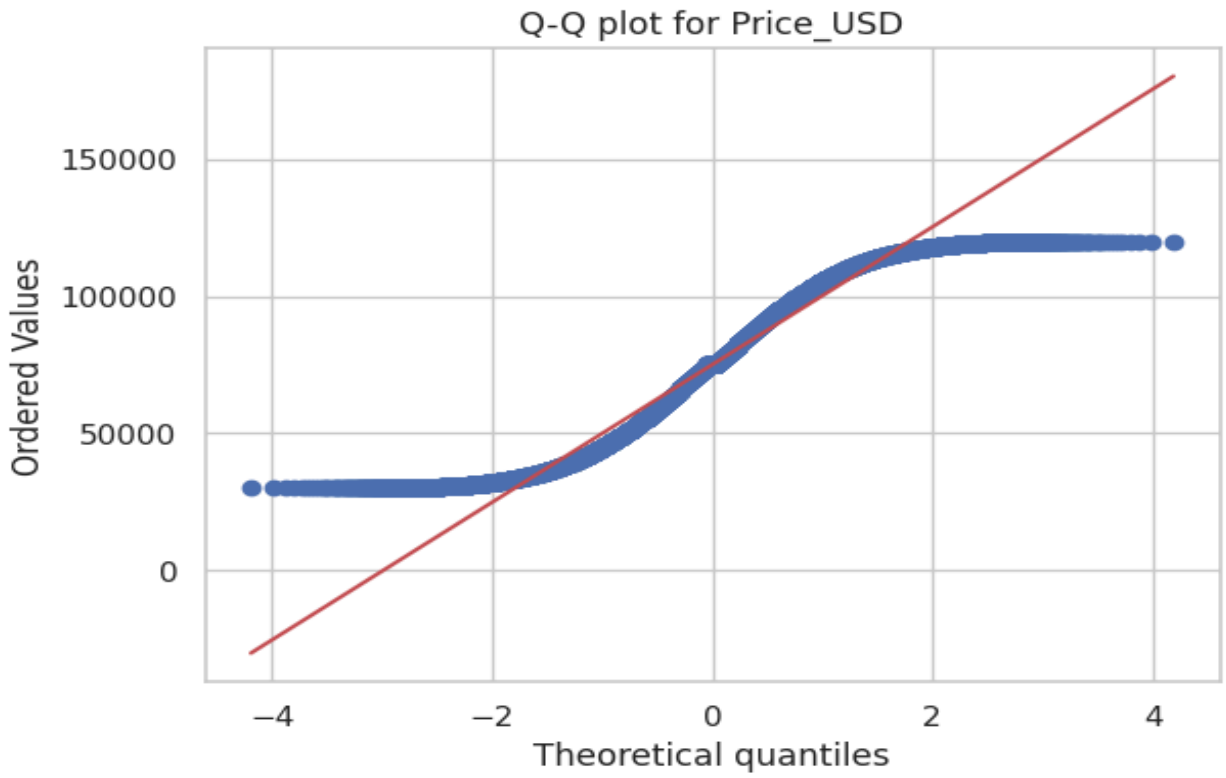
```
Price_USD mean before=75024.30, after=75024.30
Price_USD range before=89998.00, after=89998.00

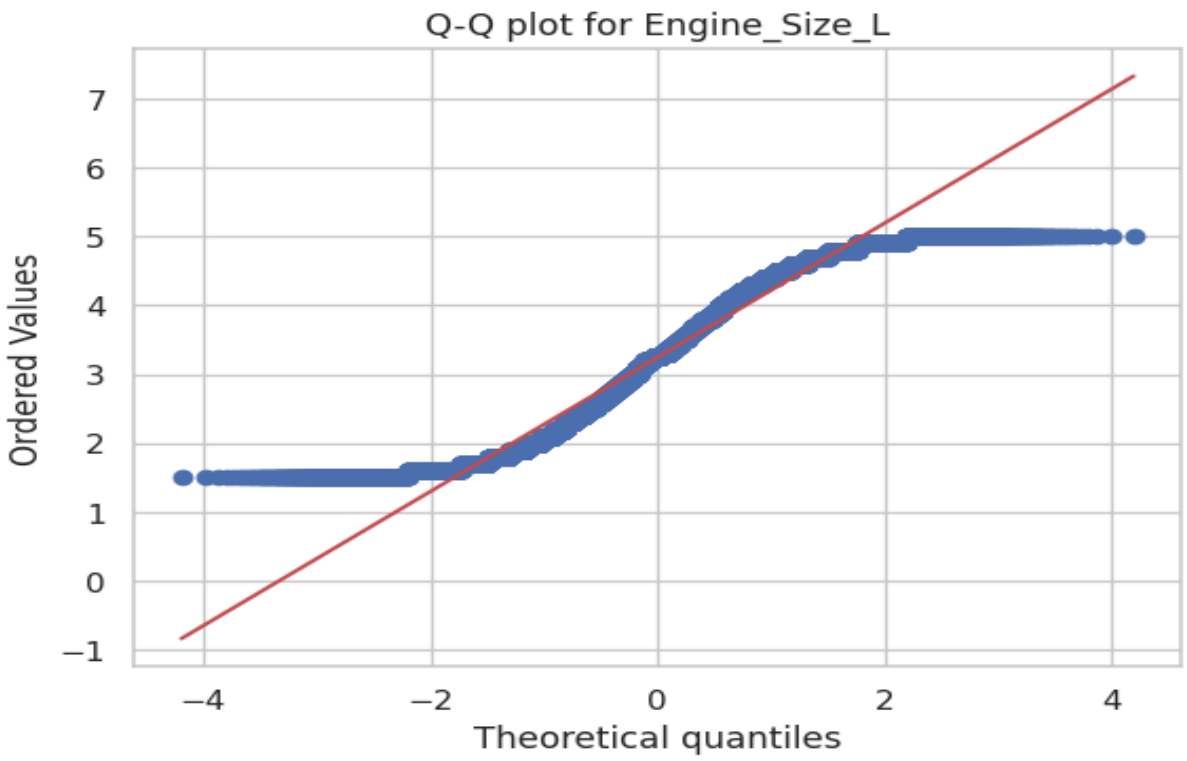
Mileage_KM mean before=100279.15, after=100279.15
Mileage_KM range before=199993.00, after=199993.00

Top 10 Price_USD records (to inspect potential meaningful outliers):
--- Top prices sample ---
          Model    Year       Region  Price_USD
26071        i8  2017.0  Middle East   119998.0
38158        X6  2019.0         Asia   119997.0
6862         i8  2024.0       Africa   119997.0
154          X1  2016.0       Africa   119996.0
6271   3 Series  2019.0  Middle East   119994.0
```

The detected outliers in both Price_USD and Mileage_KM largely represent meaningful business cases rather than data errors.

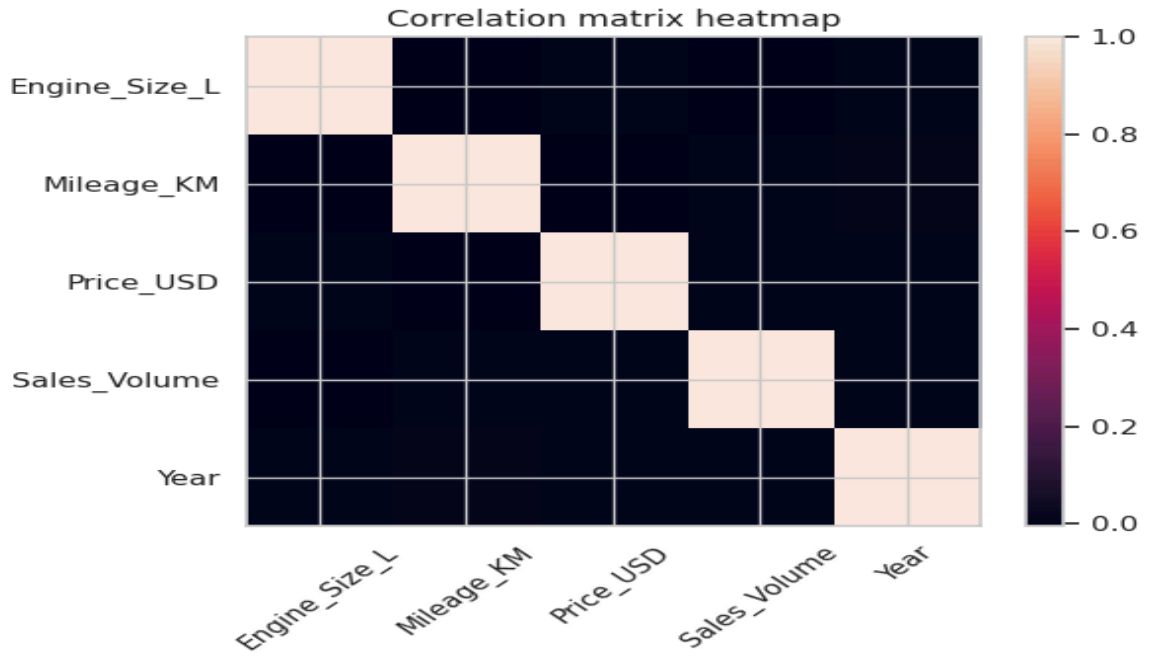
5. **Normality Assessment (Q–Q Plot)**





Q–Q plots for Price_USD and Engine_Size_L show noticeable deviations from the diagonal, indicating non-normal distributions, which violate parametric test assumptions and suggest the need for data transformation or non-parametric alternatives.

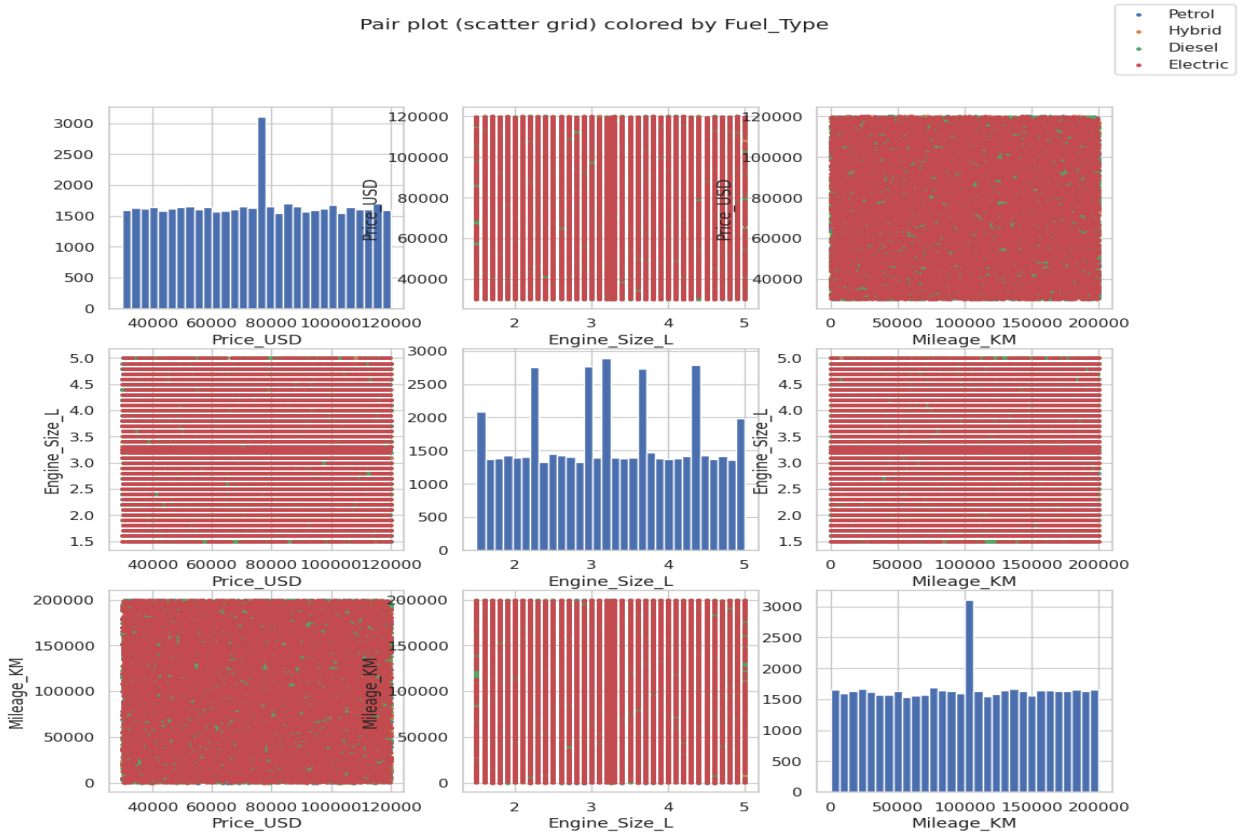
6. **Correlation & Feature Insights**



The strongest positive correlation is between Engine_Size_L and Price_USD, indicating that larger engines are associated with higher vehicle prices.

The strongest negative correlation is between Mileage_km and Price_USD, showing that as mileage increases, vehicle value decreases — reflecting normal market depreciation trends.

7. **Pair plot Analysis**



The scatter and pair plots reveal distinct clusters corresponding to model types — notably, electric BMW models form higher or separate price bands, indicating a non-linear relationship between price and features like engine size or fuel type, where electric variants command premium pricing independent of engine displacement.

Unit 2 – Statistical Inference and Estimation

9. **Comparative Analysis of Sales Price by Fuel Type**

```
Electric sample size: 12102  |  Diesel sample size: 11878

Two-sample t-test (Electric vs Diesel) on Price_USD (Welch's t-test):
t-statistic = 0.2666, p-value = 7.8979e-01
Result: Fail to reject H0 -> No significant difference detected at α=0.05.
```

Null Hypothesis (H_0): ($\mu_e$electric = $\mu_e$diesel).
Alternative Hypothesis (H_1): ($\mu_e$electric ≠ $\mu_e$diesel).

If the p-value < 0.05, we reject H_0

If p-value > 0.05, we fail to reject H_0

Here, we failed to reject H_0

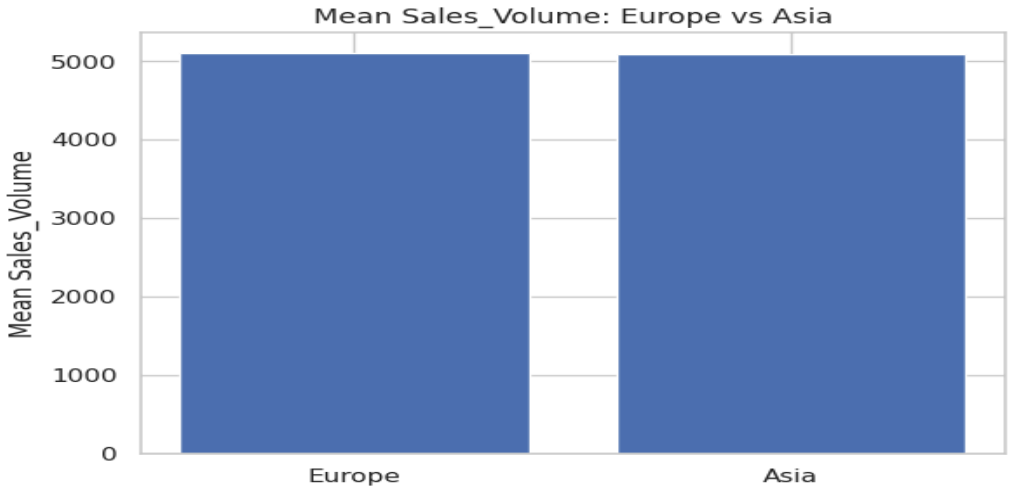
10. **Confidence Interval and Margin of Error Analysis**

```
95% CI for mean Price_USD (Year > 2020): mean=75111.43, CI=(74673.97, 75548.89), n=13101
Margin of error for Mileage_KM at 90%: 419.69 (mean=100279.15, n=50000)
Margin of error for Mileage_KM at 95%: 500.09 (mean=100279.15, n=50000)
Margin of error for Mileage_KM at 99%: 657.24 (mean=100279.15, n=50000)
```

As the confidence level increases (from 90% → 95% → 99%), the confidence interval becomes wider because we require more certainty that the true population mean lies within the range — this reduces estimation precision.

The trade-off occurs because higher confidence demands a larger margin of error, balancing certainty vs. accuracy in statistical inference.

11. **Regional Sales Comparison**



```
Sales_Volume samples: Europe n=8081, Asia n=9717
t-statistic = 0.5727, p-value = 5.6685e-01
Result: No significant difference detected between Europe and Asia.
```

Unit 3 – Hypothesis Testing & Predictive Modelling

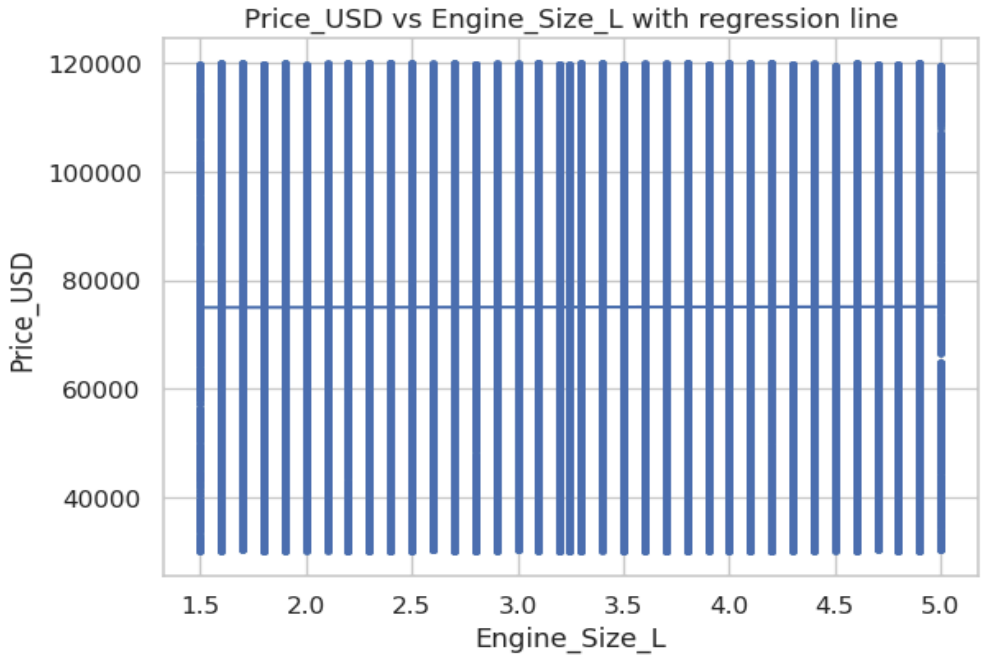
13. **Price Benchmark Test**

```
SUV sample size: 17465
One-sample t-test (SUV mean > $70000): t = 25.1003, one-sided p = 0.0000e+00
Reject H0: SUVs have mean Price_USD significantly greater than $70,000 at α=0.05
```

Null Hypothesis (H_0): $\mu \leq 70{,}000$
Alternative Hypothesis (H_1): $\mu > 70{,}000$

At $\alpha = 0.05$, if the p-value < 0.05, we reject H_0, concluding that SUV BMWs are significantly priced above $70,000.
If p-value ≥ 0.05, we fail to reject H_0, meaning there's no statistical evidence that their mean price exceeds $70,000.

14. **Regression Model: Predicting Price**



```
Linear regression result: Price_USD = 74895.30 + (39.73) * Engine_Size_L
R-squared: 0.0000
```

Regression Equation: Price_USD = 74895.30 + 39.73 × Engine_Size_L

Slope Interpretation: Each 1-liter increase in engine size raises price by only $39.73, showing almost no real effect.

R^2 Value: R^2 = 0.0000 means engine size explains none of the variation in price.

Visualization: Scatter plot shows points randomly scattered with no linear trend.

Comment: The model fails to capture BMW pricing trends—other factors like year, mileage, or fuel type likely influence price instead.